

# AthDGC - Athens Diachronic Glossa Chronos (Athens-PROIEL)

An open PROIEL-style treebank with Indo-European parallels

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## Background

Greek is the longest continuously attested member of the Indo-European family. The PROIEL Treebank (Haug & Jøhndal 2008; Eckhoff *et al.* 2018) established the gold standard for syntactically annotated parallel IE Bible corpora, but its scope is restricted to the New Testament. No openly licensed end-to-end pipeline

currently covers the full diachronic record of Greek with explicit argument-structure tagging and reproducible cross-lingual alignment.

## Approach

AthDGC (“Athens-PROIEL”) is an open end-to-end pipeline that:

1. **Discovers** new Greek + parallel-language sources daily (archive.org, Perseus, First1K, Wikisource, Diorisis)
2. **Filters** via Greek-script ratio, apparatus-criticus rejection, content-hash dedup
3. **Converts** to PROIEL XML 2.0
4. **Annotates** with Stanza `grc_proiel` (+ `la_proiel`, `cu_proiel`, `got_proiel` for parallels)
5. **Extracts** per-verb argument structure (subj, obj, oblique, voice, aspect)
6. **Aligns** cross-lingually via LaBSE (sentence) + AwesomeAlign (word, mBERT)
7. **Stores** as PROIEL XML + JSONL partitions + Qdrant vectors + Neo4j alignment graph

## Coverage (v0.4)

- 89.9 M corpus rows
- 4.08 M annotated Greek rows

- 6,861 NT-aligned Greek verses
- 4 cross-aligned IE witnesses (lat, got, chu, xcl)
- 8 diachronic periods (Archaic to Modern)
- 80+ TLG canonical authors with curated metadata

## Argument-structure capture

Per VERB / AUX token AthDGC stores:

- subject, direct object, indirect / oblique args
- voice (active / middle / passive)
- aspect (perfective / imperfective)
- tense, mood, verb-form, person, number

stored in a column queryable across the cross-lingual alignment edges.

## Example query

“Show every Greek aorist transitive verb with an accusative object whose Latin Vulgate counterpart is a passive periphrastic”

Neo4j Cypher over edges, filtering source by target by

```
(:Token)-[:TRANSLATED_AS]->(:Token)
(voice=Act,aspect=Perf,obj=Acc) and
(voice=Pass,verb_form=Periphrastic)
```

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## Open access

- Source code (Apache-2.0): [github.com/AthDGC/Diachronic-Linguistics-Platform](https://github.com/AthDGC/Diachronic-Linguistics-Platform)
- Dataset (CC-BY-4.0): [10.5281/zenodo.20439182](https://doi.org/10.5281/zenodo.20439182)
- Showcase: [athdgc.github.io](https://athdgc.github.io)

- HuggingFace: [athdgc/diachronic-greek-corpus](https://huggingface.co/athdgc/diachronic-greek-corpus) (forthcoming)